

Joydeep Banik

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Summary

Electronics and Communication Engineering undergraduate with a strong foundation in backend development, distributed systems, applied ML and cloud platforms. Hands-on experience in backend development and computer vision across end-to-end SDLC workflows. Co-authored a paper accepted at IPTA 2026. Finalist at India Innovates 2026 and semi-finalist at Smart India Hackathon, 2025.

Technical Skills

Programming Languages: Python, C++, JavaScript, TypeScript, HTML, CSS, Bash, MySQL, PostgreSQL
Backend & Framework: Django, Flask, Node.js, Next.js REST API, Supabase
Data & ML: TensorFlow, PyTorch, OpenCV, Rasterio
Cloud & DevOps: Docker, Kubernetes, Google Cloud Platform(GCP), Git, Github Actions

Internships

Research Intern May 2026 – Present
Jadavpur University

- Co-authored a study benchmarking DehazeNet, FFA-Net, and FD-GAN across 9 synthetic, real, and remote-sensing haze datasets using PSNR, SSIM, MS-SSIM, and MSE; accepted for presentation at IEEE IPTA 2026.
- Co-developing a retrieval-augmented dehazing framework that retrieves latent-space features to improve frame-level clarity and preserve temporal consistency across video sequences.

Software Engineer Intern Dec 2025 – Jan 2026
Haldia Petrochemicals Ltd.

- Co-engineered an API-driven supply chain and inventory tracking system handling end-to-end stock monitoring, procurement workflows, and warehouse logistics deployed in an active petrochemical plant using agile development methodologies.
- Built relational SQL schemas and backend validation logic to track historical inventory movement, prevent discrepancies, and provide operational visibility for logistics teams.
- Monitored enterprise network telemetry and data flow using PRTG, triaging operational incidents to maintain 99%+ infrastructure and data system uptime.

Projects

CTFy GitHub

- Engineered an event-driven tournament engine processing real-time participant submissions, instant point tallying, and dynamic podium medal assignment with minimal database load.
- Optimized Next.js 16 Turbopack build and caching pipelines, achieving sub-2s static route compilation and seamless edge deployment on Vercel with zero cold-start bottlenecks.
- Constructed end-to-end schema validation pipelines utilizing Zod and React Hook Form across server actions and client inputs, guaranteeing 100% type safety and zero unhandled schema anomalies.
- Automated user lifecycle management via PostgreSQL triggers and triggers-based profile synchronization, maintaining relational integrity across high-frequency competition events.

KubeSense GitHub

- Designed a decoupled, asynchronous engine using Python asyncio and dataclasses, orchestrating concurrent API polling, terminal UI rendering, and streaming log ingestion without thread blocking.
- Applied the Abstract Factory and Strategy design patterns to abstract multiple LLM backends (Ollama, Groq, OpenAI), enabling runtime provider hot-swapping and deterministic heuristic fallbacks.
- Achieved comprehensive test coverage using pytest and pytest-asyncio, unit testing asynchronous SSH streams, JSON sanitizers, and edge-case regex parsing across distributed log formats.
- Optimized UI rendering performance by binding Textual reactive properties to state changes, triggering partial DOM re-renders exclusively on updated widgets.

Sovereign Bharat Intelligence Graph

- Developed a distributed, fault-tolerant data ingestion pipeline using Apache Kafka to unify structured, unstructured, and real-time feeds into a Neo4j-backed intelligence graph.
- Implemented rule engines across domains for early risk detection and scenario simulation.
- Selected for on-site presentation as finalists at India Innovates, 2026.

AgriSense AI

- Built a Python pipeline with Rasterio that converts raw multi-spectral Landsat 9 GeoTIFF scenes into model-ready data through preprocessing, atmospheric correction, and stacking of spectral bands into aligned raster stacks.
- Selected at internals for Smart India Hackathon, 2025.

Education

2023 – 2027 **B.Tech in Electronics & Communication Engineering** at RCC Institute of Information Technology
2021 – 2023 **Higher Secondary** (CBSE), Kalyani Public School
2011 – 2021 **Secondary** (ICSE), St. Claret School